🥗 Anita's Veg Cafe

SQL Case Study: From Home Kitchen to Data-Driven Business

📊 Project Dashboard

Project Name	Anita's Veg Cafe — Customer & Sales Analysis
Location	Bengaluru, India
Database	`anitas_veg_cafe` (Sales, Menu, Members)
Tech Stack	PostgreSQL (Joins, CTEs, Window Functions, Case Logic)
Scope	11 Strategic Business Questions

🏗 The Data Architecture

Schema Overview

Before we dive into the insights, let's look under the hood. The database `prj` captures three distinct dimensions of the business: Products, People, and Transactions.

menu → sales ← members

Database Blueprint

1. The Menu (Products) — Stores the core offering and pricing.

```
CREATE TABLE menu (
    "product_id" INTEGER,
```

```
    "product_name" VARCHAR(50),
    "price" INTEGER
);

INSERT INTO menu VALUES
(1, 'Paneer Butter Masala', 180),
(2, 'Veg Biryani', 150),
(3, 'Masala Dosa', 120);
```

2. The Members (Loyalty) — Tracks when customers officially joined the program.

```
CREATE TABLE members (
    "customer_id" VARCHAR(10),
    "join_date" DATE
);

INSERT INTO members VALUES
('Aarav', '2021-01-07'),
('Meera', '2021-01-09');
```

3. The Sales (Transactions) — The heartbeat of the cafe—every order logged here.

```
CREATE TABLE sales (
    "customer_id" VARCHAR(10),
    "order_date" DATE,
    "product_id" INTEGER
);

INSERT INTO sales VALUES
('Aarav', '2021-01-01', 1), ('Aarav', '2021-01-01', 2),
('Aarav', '2021-01-07', 2), ('Aarav', '2021-01-10', 3),
('Aarav', '2021-01-11', 3), ('Aarav', '2021-01-11', 3),
('Meera', '2021-01-01', 2), ('Meera', '2021-01-02', 2),
('Meera', '2021-01-04', 1), ('Meera', '2021-01-11', 1),
('Meera', '2021-01-16', 3), ('Meera', '2021-02-01', 3),
('Rohan', '2021-01-01', 3), ('Rohan', '2021-01-01', 3),
('Rohan', '2021-01-07', 3);
```

🕵️‍♂️ Section I: The Customer Journey

Understanding who visits, how often, and what hooks them.

Q1. Who are the High-Value Customers?

Goal: Calculate total spend and order volume per customer.

```
SELECT
    s.customer_id,
    SUM(m.price) AS total_spending,
    COUNT(*) AS total_orders
FROM sales s
JOIN menu m ON s.product_id = m.product_id
GROUP BY s.customer_id
ORDER BY total_spending DESC;
```

📊 The Insight

Aarav is the clear VIP (highest spend), largely due to ordering premium Paneer dishes. Rohan, while frequent, sticks to the lowest-priced item (Masala Dosa).

💡 Business Move

Highlight Aarav as a "Top Spender." A simple "Thank You" note or a reserved table during peak hours costs nothing but builds immense loyalty.

Q2. How Often Do They Visit?

Goal: Count distinct visit days (not just orders) to measure habit formation.

```
SELECT
    customer_id,
    COUNT(DISTINCT order_date) AS visit_days
FROM sales
GROUP BY customer_id
ORDER BY visit_days DESC;
```

📊 The Insight

This separates "habituals" from "occasional" customers. Ordering twice in one day is loyalty; ordering twice a week is habit.

💡 Business Move

Target high-frequency customers (like Meera) with a "Buy 5 Visits, Get 1 Free" physical punch card to solidify the habit loop.

Q3. The "First Date" Dish

Goal: Identify the very first item a customer ever ordered.

```
SELECT customer_id, product_name AS first_dish
FROM (
    SELECT
        s.customer_id,
        m.product_name,
        ROW_NUMBER() OVER (
            PARTITION BY s.customer_id
            ORDER BY s.order_date, s.product_id
        ) AS rn
    FROM sales s
    JOIN menu m ON s.product_id = m.product_id
) t
WHERE rn = 1;
```

📊 The Insight

The "First Dish" is the entry point. If Aarav's first order was Paneer Butter Masala, he started as a high-value customer from Day 1.

💡 Business Move

If most newcomers start with the cheap Masala Dosa, create a "Newcomer Combo" that upsells them to Biryani or Paneer at a slight discount to raise their baseline expectation.

🍛 Section II: Menu Engineering

Analyzing product popularity and customer preferences.

Q4. The Crowd Favorite

Goal: Find the single most popular item by order volume.

```
SELECT
    m.product_name,
    COUNT(s.product_id) AS total_orders
FROM sales s
JOIN menu m ON s.product_id = m.product_id
GROUP BY s.product_id, m.product_name
ORDER BY total_orders DESC
LIMIT 1;
```

📊 The Insight

While Masala Dosa might lead in volume (due to price and breakfast appeal), Paneer Butter Masala likely drives the actual revenue.

💡 Business Move

Never run out of the volume leader. Promote a "Dosa + Chai" combo to increase the ticket size of your most frequent order.

Q5. The "Comfort Dish"

Goal: Determine the most frequently ordered item *per specific customer*.

```
SELECT customer_id, product_name, order_count
FROM (
    SELECT
        s.customer_id,
        m.product_name,
        COUNT(*) AS order_count,
        DENSE_RANK() OVER (
            PARTITION BY s.customer_id ORDER BY COUNT(*) DESC
        ) AS rnk
    FROM sales s
    JOIN menu m ON s.product_id = m.product_id
    GROUP BY s.customer_id, m.product_name
) t
WHERE rnk = 1;
```

📊 The Insight

Everyone has a "default." Knowing this allows for hyper-personalization (e.g., prepping their usual before they walk in).

💡 Business Move

Automated marketing: "Hi Aarav! Your favorite Paneer Butter Masala is ₹20 off today." Personalization drives retention better than generic blasts.

🎫 Section III: Decoding Loyalty

Analyzing the effectiveness of the membership program.

Q6. The First Member Order

Goal: What did members order immediately *after* joining?

```
WITH post_join AS (
    SELECT lc.customer_id, s.order_date, lc.join_date, m.product_name
    FROM sales s
    JOIN menu m ON s.product_id = m.product_id
    JOIN loyal_customers lc ON s.customer_id = lc.customer_id
    WHERE s.order_date >= lc.join_date
),
ranked AS (
    SELECT *,
    DENSE_RANK() OVER (PARTITION BY customer_id ORDER BY order_date) AS fs
    FROM post_join
)
SELECT customer_id, product_name, order_date
FROM ranked
WHERE fst = 1;
```

📊 The Insight

Did joining change their behavior? If they ordered a premium item immediately, the program successfully incentivized an upgrade.

💡 Business Move

Trigger a "Welcome to the Club" SMS immediately upon signup: "Earn 50 bonus points on your very next order!"

Q7. The "Conversion Meal"

Goal: What was the last thing they ate *before* deciding to join?

```
WITH pre_join AS (
    SELECT lc.customer_id, s.order_date, m.product_name
    FROM sales s
    JOIN menu m ON s.product_id = m.product_id
    JOIN loyal_customers lc ON s.customer_id = lc.customer_id
    WHERE s.order_date < lc.join_date
),
ranked AS (
    SELECT *,
    RANK() OVER (PARTITION BY customer_id ORDER BY order_date DESC) AS lst
    FROM pre_join
)
SELECT customer_id, product_name, order_date
FROM ranked
WHERE lst = 1;
```

📊 The Insight

This identifies the tipping point. What meal convinced them they loved this place enough to sign up?

💡 Business Move

If the conversion meal is Paneer, place sign-up flyers specifically on tables ordering Paneer: "Love this dish? Join us and earn double points!"

Q8. The Cost of Conversion

Goal: How much did a customer spend *before* becoming a member?

```
WITH pre_join AS (
    SELECT lc.customer_id, m.price
    FROM sales s
    JOIN menu m ON s.product_id = m.product_id
    JOIN loyal_customers lc ON s.customer_id = lc.customer_id
    WHERE s.order_date < lc.join_date
)
SELECT
    customer_id,
    COUNT(*) AS items_ordered_before_joining,
    SUM(price) AS total_spent_before_joining
FROM pre_join
GROUP BY customer_id;
```

📊 The Insight

High spend before joining means you captured a loyalist late; low spend means you captured them early.

💡 Business Move

If it takes 5+ visits to convert, you're waiting too long. Train staff to mention the loyalty program at the *very first* interaction.

Q9. Points Mechanics (The Premium Nudge)

Goal: Calculate points (₹1 = 10pts), but Paneer earns double (20pts).

```
SELECT
    s.customer_id,
    SUM(
        CASE
            WHEN m.product_name = 'Paneer Butter Masala' THEN m.price * 20
            ELSE m.price * 10
        END
    ) AS loyalty_points
FROM sales s
JOIN menu m ON s.product_id = m.product_id
GROUP BY s.customer_id
ORDER BY loyalty_points DESC;
```

📊 The Insight

This logic gamifies the upsell. Customers realize ordering Paneer accelerates their rewards faster.

💡 Business Move

Menu Board Psychology: Add a "🔥 Earn DOUBLE Points!" badge next to the Paneer Butter Masala.

Q10. The "First Week" Bump

Goal: Calculate points for Aarav/Meera, assuming the first 6 days post-join earn double points on *everything*.

```
SELECT
    s.customer_id,
    SUM(
        CASE
            WHEN s.order_date BETWEEN lc.join_date AND (lc.join_date + INT
            THEN m.price * 20
            ELSE m.price * 10
        END
    ) AS total_points_jan
FROM sales s
JOIN menu m ON s.product_id = m.product_id
JOIN loyal_customers lc ON s.customer_id = lc.customer_id
WHERE s.order_date <= '2021-01-31'
GROUP BY s.customer_id;
```

📊 The Insight

This query validates if a "First Week Bonus" drives rapid engagement. It creates urgency to return immediately.

💡 Business Move

Marketing Copy: "Welcome! Double points for your first 7 days." This drives 2-3 extra visits in week one.

⚠ Section IV: Risk Management

Identifying churn before it happens.

Q11. The Churn Alert

Goal: Find customers who haven't ordered in the last 30 days.

```
SELECT DISTINCT
    s.customer_id,
    MAX(s.order_date) AS last_order_date,
    CURRENT_DATE - MAX(s.order_date) AS days_since_last_order
FROM sales s
GROUP BY s.customer_id
HAVING MAX(s.order_date) < CURRENT_DATE - INTERVAL '30 days'
ORDER BY days_since_last_order DESC;
```

📊 The Insight

Silence is expensive. These customers were active but have slipped away. The "Days Since" metric prioritizes who to save first.

💡 Business Move

The "We Miss You" Campaign: Send a WhatsApp with a 20% off code or a "Bonus Points on your next visit" offer. Re-acquisition is cheaper than acquisition.

⚠ **Design Note:** The analysis uses `LEFT JOIN` logic for loyalty data. This ensures that non-members like Rohan (who is a great customer!) are not accidentally excluded from sales calculations.

🏆 Executive Summary & Strategic Roadmap

The Big Picture: Anita's café has solid demand but distinct customer segments. The data reveals a clear path to higher revenue: Move customers from low-volume (Dosa) to high-value (Paneer) transactions using the loyalty program as the lever.

#	Finding	Strategic Action
1	**High-Value Concentration** Aarav accounts for disproportionate spend.	Launch a **VIP Tier** with exclusive perks to retain top spenders.

#	Finding	Strategic Action
2	**Volume vs. Value** Dosa drives traffic; Paneer drives profit.	Create **"Upgrade Combos"** and keep double points on Paneer.
3	**Loyalty Effectiveness** First-week double points drive returns.	Aggressively market the **"First Week Bonus'** at the signup moment.
4	**Non-Member Opportunity** Rohan buys often but isn't a member.	Target non-members with a **"Join & Save"** limited-time offer.
5	**Churn Risk** 30+ days inactivity signals loss.	Automate **"Win-Back"** SMS campaigns with discount codes.

📝 Conclusion

This project illustrates that raw data is useless without interpretation. By applying SQL from simple Joins to complex Window Functions—we transformed transaction logs inte narrative about Anita's customers and her menu.

The Takeaway: Anita no longer needs to guess. She knows who her VIPs are, wh dishes to push, and exactly when a customer is about to leave. With these insights, s can stop running her cafe on intuition and start running it on intelligence.